

Can AI Actually Make Ghanaians More Tech-Literate — Or Just More Dependent?

Evaluating AI's role in public technology literacy in Ghana

By [Your Name]

Ask most Ghanaians whether they “use technology,” and the answer is almost certainly yes. Mobile money, WhatsApp, YouTube tutorials, chatbots on customer-service lines — digital tools are woven into daily life from Accra’s markets to villages in the Upper West Region. But ask a harder question — can they tell when a message is an AI-generated scam, understand why an app is asking for their Ghana Card number, or verify a suspicious claim before sharing it — and the picture gets murkier.

That gap, between simply *using* technology and truly *understanding* it, is the subject of this project: an evaluation of whether artificial intelligence is actually making Ghanaians more capable, critical, and safe digital citizens — or whether it’s just making digital services easier to consume without making people any wiser about them.

The central question driving this research is simple to ask and hard to answer:

Does AI help Ghanaians become more capable, critical, safe, and independent technology users — or does it mainly make digital services easier to consume without improving understanding?

This question matters right now. Ghana has made real progress on digital adoption — internet penetration climbed from roughly 8% of the population in 2010 to about 69% in 2021 — but that progress is uneven. Uptake sits around 80% in urban areas versus roughly 54% in rural ones, and gaps in affordability, infrastructure, education, and language inclusion remain wide (World Bank, 2023).

Setting the Ghanaian scene

A mobile-first digital transformation

For most Ghanaians, the phone *is* the internet. That has created enormous opportunities for mobile-first AI learning tools — chatbots, audio lessons, local-language voice assistants — but a phone-first environment doesn’t automatically build the deeper digital skills people need: document creation, structured research, privacy management, or the ability to critically evaluate what an AI tool tells them.

Digital inclusion in Ghana still varies sharply by:

- Urban versus rural location
- Household income

- Age (young people versus older adults)
- Level of formal education
- Disability status
- Language — English versus Twi, Ga, Ewe, Dagbani, and other local languages

A study using Ghana’s 2021 Population and Housing Census found that only about 53% of older adults had used the internet in the preceding three months, mostly via mobile phones, with internet use strongly tied to age, education, place of residence, household wealth, and disability status (Ghana Statistical Service / *Nature Humanities and Social Sciences Communications*, 2024). Any AI-literacy programme that assumes a level playing field — same access, same confidence, same data budget, same language comfort — will fail the people who need it most.

A policy window worth using

Ghana launched its first **National Artificial Intelligence Strategy** on 24 April 2026 — a ten-year national blueprint covering education, infrastructure, data governance, research, talent development, public-sector transformation, and responsible AI use (Ministry of Communication, Digital Technology and Innovations, 2026). That gives this project real policy relevance: an evaluation of public AI literacy can help show whether these national ambitions are actually reaching ordinary citizens.

The legal groundwork already exists in pieces — the Data Protection Act (2012), the Right to Information Act (2019), and electronic transactions legislation — but UNESCO notes that Ghana still lacks comprehensive AI-specific legislation, a dedicated AI procurement regime, and a fully operational framework for algorithmic transparency, redress, and liability (UNESCO, 2026). That’s precisely why public literacy isn’t a “nice to have.” If citizens can’t recognise when they’re dealing with an automated system, understand its limits, or know how to complain about an unfair outcome, legal protections remain theoretical.

Where AI can genuinely help — and where it can hurt

Area	Potential AI contribution	Example in Ghana	Main risk
Personalised learning	Adapts explanations and feedback to the learner’s level	A chatbot walks someone through email, mobile banking, or e-government step by step	Learners accept wrong explanations without checking them
Language accessibility	Translates content, supports voice interaction	AI support in Twi, Ga, Ewe, Dagbani, and other local languages	Weak or inaccurate translation excludes users
Digital public services	Helps citizens navigate forms, tax filing, business registration	An assistant explains what documents are needed for a service	Bad advice causes missed deadlines or wrong submissions

Area	Potential AI contribution	Example in Ghana	Main risk
Media literacy	Teaches fact-checking and source verification	Training people to check suspicious videos and messages	AI itself can generate convincing misinformation
Cybersecurity awareness	Teaches safe passwords, phishing detection, scam prevention	Simulated phishing exercises in community workshops	Scams are getting more sophisticated, faster than training can keep up

AI is at its most valuable when it acts as a **learning assistant, not an answer machine**. Instead of simply telling someone whether a WhatsApp message is fraudulent, a good literacy programme teaches them to check the sender, inspect the link, spot pressure tactics, never share a one-time PIN, and verify the claim through a trusted channel.

That skill is becoming urgent. Ghana's Cybersecurity Industry Forum warned in September 2026 that criminals are already using AI to produce far more polished, convincing phishing messages, making fraud harder for both citizens and institutions to spot (MyJoyOnline, 2026). Ghana's National Cyber Security Awareness Month has, in turn, put growing emphasis on misinformation, disinformation, and the responsible use of AI and emerging technologies (Cyber Security Authority, 2025).

The real research problem

AI adoption is outpacing public understanding. Someone can know *how* to use ChatGPT, a mobile-money app, or a customer-service chatbot without knowing:

- Whether the tool's answers are actually accurate
- Whether it's collecting or storing their personal data
- How algorithmic recommendations shape what they see online
- How to spot an AI-generated image, video, or voice message
- When *not* to share sensitive information with an AI system
- How to use AI ethically at school, at work, or in business
- How to challenge, or request human review of, an automated decision

That's the line between **digital use** and **digital literacy**. A person who can open an app is a technology *user*. A person who can use it safely, judge its output, protect their data, and know its limits is technologically *literate*. This project is about measuring which one AI is actually producing in Ghana.

Aim and objectives

Aim: To evaluate the role of artificial intelligence in improving public technology literacy in Ghana, with attention to access, skills, critical thinking, safety, inclusion, and responsible use.

Objectives:

1. Assess public awareness and use of AI-enabled tools in Ghana.
2. Measure whether AI-assisted learning improves practical digital skills.
3. Examine how AI affects citizens' ability to assess information critically.
4. Identify barriers that keep disadvantaged groups from benefiting from AI-literacy programmes.
5. Evaluate public understanding of AI-related privacy, cybersecurity, bias, and misinformation risks.
6. Compare AI-supported training against conventional digital-literacy training.
7. Develop recommendations for government, schools, civil society, libraries, community centres, and technology providers.

Research questions

1. What level of awareness do Ghanaian citizens have of AI and AI-enabled services?
2. Which groups are most likely to use AI tools, and which are excluded?
3. Does using AI improve citizens' ability to perform practical digital tasks?
4. Does AI-supported learning improve media literacy and fact-checking behaviour?
5. Can citizens recognise the risks of deepfakes, phishing, automated scams, privacy breaches, and biased AI outputs?
6. How do language, location, income, gender, age, disability, education, device ownership, and data cost affect AI literacy?
7. What types of AI learning tools work best for Ghanaian communities?
8. How can Ghana build an inclusive, ethical, and locally relevant model of AI literacy?

A five-dimensional framework for measuring AI literacy

Rather than treating “AI literacy” as one vague idea, this project breaks it into five measurable dimensions.

1. Access — Can people actually reach and use AI-enabled tools? This covers device and internet availability, data and electricity costs, local-language interfaces, accessibility for people with disabilities, and gender, age, and rural-urban participation gaps. A programme that only reaches urban, English-speaking, already-connected users cannot be called a success.

2. Functional skills — Can participants complete basic digital tasks: managing an email account, finding reliable information online, using e-government forms, making mobile-money transactions securely, or using AI to summarise information and then checking that summary?

3. Critical AI literacy — Do people understand that AI output is not automatically true, neutral, private, or fair? Can they ask: *Who built this tool? What data might it rely on? Can it be wrong? Does it show sources? Could it be unfair to some people? Is it safe to share this information? Should I get a human opinion instead?*

4. Safety and ethical use — Can people protect themselves and others? This spans password security, phishing recognition, safe data handling, spotting synthetic media,

avoiding academic dishonesty, and understanding consent before uploading someone else's photo, voice, or data to an AI system.

5. Empowerment and outcomes — The strongest proof of literacy isn't a quiz score; it's whether people gain real independence. Better access to government or financial services, improved job prospects, more confidence using digital tools, less vulnerability to fraud, and stronger participation in civic and business life online.

How to actually measure it: a mixed-methods approach

A **quasi-experimental pre-test/post-test design** offers the strongest evidence here, comparing two groups:

- **Intervention group** — receives AI-supported digital-literacy training
- **Comparison group** — receives standard digital-literacy training without AI tools, or receives the AI programme later

This structure makes it possible to credit (or blame) AI specifically, rather than crediting “training in general.”

Study locations should reflect Ghana's diversity rather than treating it as one uniform digital environment — Greater Accra or Kumasi for urban participation, a peri-urban municipality, a rural community in the northern, middle, or coastal belt, and a community with limited connectivity. The digital-literacy needs of a university student in Accra, a market trader in Kumasi, a teacher in Tamale, and a farmer in a rural district are simply not the same.

Participants could be drawn from secondary and tertiary students, teachers, small-business owners and informal-sector workers, public servants, market traders, parents and caregivers, older adults, persons with disabilities, and rural or local-language-speaking residents. A sample of roughly 300–600 participants across these locations would produce solid quantitative evidence for a study of this scope.

Data collection would combine:

- **A survey** covering demographics, device and internet access, AI use frequency, basic digital skills, understanding of AI concepts, perceived benefits and fears, privacy practices, and confidence using online services.
- **A practical skills assessment**, which matters more than self-reported confidence — for example, asking participants to search for a public service online, judge whether a website is official, ask an AI tool to explain something and then identify at least two possible errors in its answer, detect a phishing message, or cross-check an AI-generated claim against a reliable source.
- **Focus groups**, to surface what surveys miss: fear of job loss to AI, trust (or distrust) of government and platforms, local-language barriers, worries about cheating in school, past experiences with scams, and gendered barriers to owning or using a device.

- **Interviews** with teachers, ICT trainers, cybersecurity professionals, local-government officials, civil-society representatives, community leaders, and — where feasible — staff from the National Information Technology Agency, the Data Protection Commission, or the Cyber Security Authority.
- **Direct observation** of training sessions, checking things surveys can't capture: Is the connection stable? Are instructions in English only? Are participants sharing devices? Are trainers explaining AI's limitations, or just its features? Are women, older adults, and people with disabilities participating equally?

A sample intervention: six weeks, community-based

Week	Topic	Practical outcome
1	Digital foundations	Device basics, internet navigation, account security
2	Understanding AI	What AI is, where it's used, and what it can't reliably do
3	AI for learning and work	Using AI for CV writing, basic research, translation, business communication
4	Information verification	Comparing AI answers against credible sources; spotting misinformation
5	Privacy and cybersecurity	Identifying phishing, unsafe links, fraud, and risky data sharing
6	Responsible AI use	Discussing bias, fairness, consent, intellectual property, and community impact

For this to work in a Ghanaian context, delivery has to be practical, not lecture-based: short, mobile-first lessons; low-data or offline resources where possible; local-language explanation with peer support; and delivery through trusted community spaces — schools, libraries, churches, mosques, youth centres, market associations, or district assemblies — using examples drawn from daily life, not generic global case studies.

The opportunity already exists

There's early evidence this can work. Africa ICT Right's AI-literacy programme in Ghana reported delivering AI and digital-pedagogy training to 29 District STEM Coordinators and 466 STEM teacher educators across seven three-day sessions since launching in 2025 (Team4Tech / Africa ICT Right, 2026). That shows teacher capacity-building is a workable entry point — but public literacy has to extend well beyond formal classrooms to reach adults who are already out of school.

Ghana has also begun introducing programming, robotics, algorithms, and AI concepts into school curricula, and researchers are actively studying teacher readiness for AI education at the Junior High School level (Computing Education Research, n.d.). That creates a natural bridge: students who learn AI concepts in school can help their own families and communities build the same understanding at home.

The risks this project has to take seriously

Digital inequality. If quality AI tools require expensive devices, reliable electricity, paid subscriptions, or English fluency, AI can widen — not close — the digital divide. A project that only reaches digitally confident people will produce results that look better than reality.

Misinformation and deepfakes. AI-generated text, audio, image, and video content is getting harder to distinguish from the real thing. Ghana's Cyber Security Authority has already increased its focus on deepfakes and online misinformation (Ghana Fact, 2025). Literacy training should build lasting verification *habits* — checking sources, cross-referencing claims — rather than teaching narrow “detection tricks” that synthetic media will quickly outpace.

Privacy and data protection. Without clear guidance, people will enter names, ID numbers, health details, passwords, and confidential business or school information into public AI tools without understanding what happens to it afterward. Training needs to be explicit about what should never be typed into a public AI system.

Language and cultural relevance. English-only AI education excludes large parts of the population. Local languages can't be treated as an afterthought — they need to shape the examples, the trainers, the communication channels, and even how “verification” is taught, since communities evaluate information differently depending on how they typically share it.

Overdependence. If people start treating AI as an unquestioned authority, it can quietly erode independent reasoning. The core message of any literacy programme should be: *use AI to support your judgement, not replace it.*

Bias and unfair decisions. AI systems can reproduce the biases baked into their training data and design choices. Citizens need to understand that automated systems can affect hiring, credit, insurance, education, policing, welfare, and access to public services — and they need a real route to human review when an automated decision seems unfair.

Measuring success

Quantitative data can be analysed with descriptive statistics, pre/post comparisons between intervention and comparison groups, cross-tabulation by age, gender, education, location, disability, and income, and regression analysis to identify what predicts higher AI literacy. Qualitative data from interviews and focus groups can be analysed thematically, watching for recurring themes like trust, fear, language barriers, privacy, scams, employment, and community support.

Concrete indicators of success might include the share of participants who can:

- Accurately define AI in plain language
- Recognise that AI can produce false information
- Identify a phishing attempt

- Know not to share passwords, Ghana Card details, or financial credentials with a public AI tool
- Verify an AI-generated claim using an independent, credible source
- Report safer online behaviour after training

— alongside the *change* in average practical-skills scores, and the size of the gap in outcomes between urban and rural participants, and across age, gender, disability, income, and education levels.

A useful working definition: someone could be called “**AI-literate**” if they can explain one useful and one risky use of AI, use an AI tool to complete a simple task, verify an AI-generated answer against another source, identify a privacy risk, recognise a likely AI-enabled scam, and know when to seek human advice instead of relying on the tool.

Recommendations

For government: - Build AI literacy into national digital-inclusion programmes broadly — not just university or specialist tech initiatives. - Fund training specifically targeted at rural, low-income, older-adult, and disability-inclusive audiences. - Require plain-language disclosure whenever AI is used in a government service. - Establish real channels for human review, complaints, and correction of harmful automated decisions. - Invest in Ghanaian-language AI tools and locally relevant content.

For schools and universities: - Teach students how AI works, where it fails, and how to verify its outputs — not just how to use it. - Train teachers to use AI ethically for planning and feedback. - Replace blanket bans with clear, practical guidance on acceptable AI use. - Build assignments that require students to critique AI output, not just generate it.

For communities and civil society: - Deliver training through institutions people already trust — churches, mosques, market associations, youth groups. - Use community radio, local-language video, and WhatsApp learning groups as core channels, not afterthoughts. - Partner with fact-checking organisations and cybersecurity trainers. - Train peer educators who can sustain support once formal sessions end.

For technology providers: - Invest properly in Ghanaian-language support, not just token translation. - Build low-data, mobile-friendly tools by default. - Explain privacy practices in plain, non-legal language. - Make it genuinely easy to report scams, harmful content, or wrong automated decisions.

The bottom line

AI has real potential to improve public technology literacy in Ghana — by personalising learning, breaking language barriers, and making digital public services more navigable. But the right measure of success was never going to be how many people open a chatbot. It’s whether AI is producing citizens who are safer, more critical, more independent, and more empowered — not just more dependent on a tool they don’t fully understand.

Ghana's new National AI Strategy opens a real policy window for this. If AI education is built around the problems people actually face — safe mobile-money use, navigating public services, finding work, growing a small business, spotting a suspicious message, avoiding a scam — it has a genuine chance to close the digital divide instead of quietly widening it.

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A full reference list, including additional academic and policy sources on AI literacy, education, and digital inclusion, is available on request.